Status	Finished
Started	Monday, 26 October 2026, 4:30 PM
Completed	Monday, 26 October 2026, 4:35 PM
Duration	4 mins 45 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Insertion

c. Complexity

d. Bubble ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓

b. It can only be applied to large lists

c. It is slower than linear search for small lists

d. It does not work with negative numbers

Question **3**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. Item is the last element in the array or item is not there at all

c. When the item is somewhere in the middle of the array ✓

d. When the item is the last element in the array

Question **4**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Hash search
- c. Binary search
- d. Both (1) & (3)

Question **5**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. The first element is returned
- b. An error is raised
- c. The last element is returned
- d. The search is considered unsuccessful ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Dividing the list in half repeatedly
- c. Checking each element sequentially ✓
- d. Skipping every second element

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 89 and 94
- b. 90 and 99 ✓
- c. 90 and 100
- d. 94 and 99

Question **8**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. At the beginning of the list
c. In the left sublist ✓
d. In the entire list

Question **9**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O👎
b. O(log n)
c. O(1) ✓
d. O(n log n)

Question **10**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. At the beginning of the list
c. In the middle of the list
d. In the left sublist

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The list is sorted
c. The search stops
d. The search continues in the left or right sublist ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. 1
c. log n
d. n ✓

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 100
c. 10 ✓
d. 1000

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By comparing each element sequentially
b. By using a hash function
c. By starting from the first element
d. By dividing the list length by two ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓
b. O👎
c. O(n log n)
d. O(1)